



Evaluation of the α Syn seeding capacity in FFPE skin biopsies of patients with Parkinson's

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Summary (or Abstract)

Recent studies have detected α -synuclein (α Syn) seeding in formaldehyde-fixed paraffin-embedded (FFPE) tissues from patients with synucleinopathies using seed amplification assays (SAAs). However, the sensitivity has been relatively low due to limited protein extraction efficiency. Aiming to provide an alternative to frozen tissues, we developed a streamlined protein extraction protocol to evaluate disease-specific seeding in FFPE human brain tissue. Using this method, our team demonstrated that proteins extracted from FFPE samples preserve their seeding potential and exhibit disease-specific seeding in Parkinson's disease and multiple system atrophy. After successfully testing this method on human brain tissue, we investigated its applicability to other peripheral fixed tissues that could serve as biomarkers. For this purpose, we extracted proteins from FFPE skin samples from 35 subjects who participated in the Parkinson's Progression Markers Initiative (PPMI). We then performed SAAs to assess the presence of misfolded α Syn in the protein homogenates derived from FFPE skin samples.

Method

For the protein extraction, the FFPE Tissue Dissociation Kit (Miltenyi BioTec, Auburn, CA, USA) was used for antigen retrieval and dissociation according to the manufacturer's instructions with slight modifications. In brief, deparaffinized tissues were transferred to C-tubes and, with the provided buffer, incubated in an 80 °C water bath for 75 min. Once the samples were cooled, the supernatant was carefully removed, and with the provided buffer and enzymes, the samples were dissociated on the gentleMACS™ Dissociator (Miltenyi BioTec, Auburn, CA, USA). The dissociated tissues were then transferred into 1.5 mL low protein binding tubes and centrifuged at 4 °C for 5 min at 5000× *g*. The supernatant was discarded, and the tissue pellet was resuspended in PBS. The tubes were centrifuged once more, at 4 °C for 5 min at 5000× *g*, and the supernatant was removed. The resulting tissue pellet was weighed and resuspended in ice-cold 10% w/v PBS before being transferred to homogenizing tubes (Precellys® 2 mL Protein Safe Soft Tissue





Homogenizing tubes containing zirconium oxide beads (CK14; Bertin Corp, Rockville, MD, USA). Following homogenization, the tissue homogenate was transferred to 1.5 mL low protein binding tubes, incubated for 10 min on ice, and centrifuged for 10 min at 4 °C 10,000× *g*. Then, the supernatant was stored at -80 °C. Then, αSyn SAA was performed. In brief, the SAA reaction mixture was composed of 40 mM phosphate buffer (pH 8.0), 350 mM Na3Citrate (Sigma, St. Louis, MI, USA), 0.1 mg/mL K23Q human recombinant αSyn (Impact Biologicals, PA, USA), 5µl of protein homogenate and 10 µM ThT (Sigma, St. Louis, MI, USA). Recombinant αSyn was thawed from -80 °C storage and filtered through a 100 kDa Amicon ultra-0.5 centrifugal filter unit (Sigma, St. Louis, MI, USA). All statistical analyses were performed using Prism (v.10, GraphPad Software, San Diego, CA, USA).

References

1. Kim A, Martinez-Valbuena I, Li J, Lang AE, Kovacs GG. Disease-Specific α-Synuclein Seeding in Lewy Body Disease and Multiple System Atrophy Are Preserved in Formaldehyde-Fixed Paraffin-Embedded Human Brain. *Biomolecules*. 2023 Jun 2;13(6):936. doi: 10.3390/biom13060936. PMID: 37371515; PMCID: PMC10296376.

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